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import numpy as np

import matplotlib.pyplot as plt

from sklearn import datasets

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.svm import SVC

from sklearn.metrics import accuracy_score, classification_report


# Load the Iris dataset

iris = datasets.load_iris()

print("Dataset Features:", iris.feature_names)

X = iris.data # Features (sepal length, sepal width, petal length, petal width)

y = iris.target # Class labels (Setosa, Versicolor, Virginica)

#print("dataset", iris)

print("data", X)

print("target",y)


# Split data into training and testing sets (80% train, 20% test)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)


# Standardize the features

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)

X_test = scaler.transform(X_test)


# Train an SVM model (using RBF kernel)

svm_model = SVC(kernel='rbf', C=1.0, gamma='scale')

svm_model.fit(X_train, y_train)


# Make predictions

y_pred = svm_model.predict(X_test)
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# Evaluate the model

accuracy = accuracy_score(y_test, y_pred)
print(f"Model Accuracy: {accuracy * 100:.2f}%")
print("Classification Report:")
print(classification_report(y_test, y_pred, target_names=iris.target_names))


# Classify new samples
new_samples = np.array([[0.7, 0.5, 0.4, 0.2], [4.8, 3.4, 1.6, 0.2]])
new_samples_scaled = scaler.transform(new_samples)
predictions = svm_model.predict(new_samples_scaled)

print("New Samples Classification:")
for sample, pred in zip(new_samples, predictions):
    print(f"Sample {sample} --> Predicted class: {iris.target_names[pred]}")
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